

$$\begin{aligned} \text{Ex. 9. } \int_0^{\pi/2} \sqrt{\tan x} dx &= \int_0^{\pi/2} \sin^{1/2} x \cos^{-1/2} x dx = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{2} \cdot \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(1 - \frac{1}{4}\right)}{1} \\ &= \frac{1}{2} \pi \operatorname{cosec} \frac{\pi}{4} = \frac{\pi}{\sqrt{2}}. \end{aligned}$$

$$\begin{aligned} \text{Ex. 10. } \int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} dx &= \int_0^{\pi/2} \sin^{-1/2} x \cos^0 x dx \times \int_0^{\pi/2} \sin^{1/2} x \cos^0 x dx \\ &= \frac{1}{2} \cdot \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)} \times \frac{1}{2} \cdot \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{5}{4}\right)} = \frac{\pi}{4} \cdot \frac{\Gamma\left(\frac{1}{4}\right)}{\frac{1}{4} \Gamma\left(\frac{1}{4}\right)} = \pi. \end{aligned}$$

$$\begin{aligned} \text{Ex. 11. } \int_0^\infty \frac{x^{m-1}}{(a+bx)^{m+n}} dx &= \int_0^\infty \frac{a^{m-1}}{b^{m-1}} z^{m-1} \cdot \frac{1}{a^{m+n}(1+z)^{m+n}} \cdot \frac{a}{b} dz \text{ by the substitution } bx = az, \text{ and} \\ \text{where } m, n, a, b > 0. \end{aligned}$$

$$= \frac{1}{a^n b^m} \int_0^\infty \frac{z^{m-1}}{(1+z)^{m+n}} dz = \frac{1}{a^n b^m} B(m, n), \text{ by relation 7.} \quad [\text{C.H. 1994}]$$

$$\text{Ex. 12. } \int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{(b+cx)^{m+n}} dx; m, n, b, (b+c) > 0. \text{ Put } x = \frac{1}{1+t}.$$

$$I = \int_0^\infty \frac{t^{n-1}}{\{(b+c)+bt\}^{m+n}} dt = \frac{1}{b^n(b+c)^m} B(m, n), \text{ by Ex. 11.} \quad [\text{C.H. 2001}]$$

$$\text{Ex. 13. } \int_0^{\pi/2} \frac{\sin^{2m-1} \theta \cos^{2n-1} \theta d\theta}{(a \sin^2 \theta + b \cos^2 \theta)^{m+n}}; m, n > 0. \text{ Put } bx = a \tan^2 \theta.$$

$$\begin{aligned} I &= \int_0^\infty \frac{\sin^{2m-1} \theta \cos^{2n-1} \theta}{(1+x)^{m+n} \cdot b^{m+n} \cdot \cos^{2(m+n)} \theta} \cdot \frac{b \cos^3 \theta}{2a \sin \theta} dx = \frac{1}{2a^m b^n} \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}} dx \\ &= \frac{1}{2a^m b^n} B(m, n), \text{ from relation 7.} \end{aligned}$$

Examples XIV

$$1. \text{ Show that } \Gamma(6) = 120, \Gamma\left(\frac{5}{2}\right) = \frac{3}{4} \sqrt{\pi}, \Gamma\left(\frac{9}{2}\right) = \frac{105}{16} \sqrt{\pi}.$$

$$2. \text{ Defining } \Gamma(x) \text{ by } \Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt, 0 < x < \infty, \text{ prove that } \Gamma(x) \text{ converges for any real } x > 0.$$

[C.H. 1996]

and that

$$(a) \Gamma(x) \text{ is continuous and differentiable on } 0 < x < \infty.$$

$$(b) \text{ For any } a > 0, \Gamma(x) = a^x \int_0^\infty e^{-at} t^{x-1} dt.$$

[C.H. 1990]

$$(c) \Gamma(x+1) = x\Gamma(x), x > 0.$$

$$(d) \Gamma(n+1) = n!, n = 1, 2, 3, \dots$$

$$(e) \text{ For all integers } n \geq 1 \text{ and all } x > 0, \Gamma(x+n) = (x+n-1) \cdots (x+1)x\Gamma(x).$$

(f) Show that $\Gamma(x) > \frac{1}{e} \int_0^1 t^{x-1} dt = \frac{1}{e^x}$, $x > 0$.

[C.H. 1998]

(g) $\Gamma(0+) = \lim_{x \rightarrow 0+} \Gamma(x) = \infty$.

(h) $\Gamma(1) = 1$.

3. Defining $B(x, y)$ by $B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$, $x, y > 0$, prove that $B(x, y)$ converges for every real $x, y > 0$.

[C.H. 2006]

Also prove

(a) $B(x, y)$ is continuous for $x > 0$, $y > 0$

(b) $B(x, y) = \int_0^\infty \frac{t^{x-1}}{(1+t)^{x+y}} dt = \int_0^\infty \frac{t^{y-1}}{(1+t)^{x+y}} dt = B(y, x)$, $x, y > 0$

[C.H. 1999]

(c) $B(x, y) = B(y, x)$.

[C.H. 1997]

(d) $B(x, 1) = \frac{1}{x}$

(e) $B\left(\frac{1}{2}, \frac{1}{2}\right) = \pi = \int_0^\infty \frac{dt}{\sqrt{t(1+t)}}$

(f) $B(x, y) = 2 \int_0^{\pi/2} \sin^{2x-1} \theta \cos^{2y-1} \theta d\theta$

[C.H. 1995]

(g) $B(x, y) = \int_0^1 \frac{t^{x-1} + t^{y-1}}{(1+t)^{x+y}} dt = \frac{1}{2} \int_0^\infty \frac{t^{x-1} + t^{y-1}}{(1+t)^{x+y}} dt$

[C.H. 2005]

(h) $B(x, x) = 2^{1-2x} B\left(x, \frac{1}{2}\right)$

(i) $\int_0^n \left(1 - \frac{t}{n}\right)^n t^{x-1} dt = n^x B(x, n+1)$; $n = 1, 2, 3, \dots$

4. Define $B(x, y)$ and $\Gamma(x)$ with restrictions on x and y , if there be any, and show that

(a) For $x, y > 0$, $B(x, y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$

[B.H. 2001]

(b) $\Gamma(x) = 2 \int_0^\infty e^{-t^2} \cdot t^{2x-1} dt$, $x > 0$

(c) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

(d) $\Gamma\left(n + \frac{1}{2}\right) = (2n)! \sqrt{\pi} / 4^n \cdot n!$, $n = 1, 2, 3, \dots$

(e) $\int_0^{\pi/2} \sin^x \theta \cos^y \theta d\theta = \frac{1}{2} B\left(\frac{x+1}{2}, \frac{y+1}{2}\right) = \frac{1}{2} \frac{\Gamma\left(\frac{x+1}{2}\right) \Gamma\left(\frac{y+1}{2}\right)}{\Gamma\left(\frac{x+y+2}{2}\right)}$; $x, y > -1$.

[C.H. 1997]

(f) $B(x+1, y) = \frac{x}{x+y} B(x, y)$

$$(g) \Gamma\left(\frac{1}{2}\right)\Gamma(2x) = 2^{2x-1}\Gamma(x)\Gamma\left(x + \frac{1}{2}\right). \quad [\text{C.H. 2000}]$$

5. Substituting $x^2 = z$, show that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ and hence find $\int_{-\infty}^{+\infty} e^{-x^2} dx$. [C.H. 1992]

6. Show that

$$(a) \int_0^\infty e^{-x} x^{3/2} dx = \frac{3}{4}\sqrt{\pi} \quad (b) \int_0^\infty \sqrt{x} e^{-x^2} dx = \frac{\sqrt{\pi}}{3} \quad (c) \int_0^\infty e^{-x} x^{3/2} dx = \frac{3}{4}\sqrt{\pi}$$

[C.H. 1998]

$$(d) \int_0^\infty e^{-4x} x^{3/2} dx = \frac{3}{128}\sqrt{\pi} \quad (e) \int_0^\infty e^{-a^2 x^2} dx = \frac{1}{2a}\sqrt{\pi} \quad (f) \int_0^\infty 5^{-x^2} dx = \frac{1}{2\sqrt{\ln 5}} \cdot \sqrt{\pi}.$$

7. Prove that

$$(a) \Gamma(m)\Gamma(1-m) = B(m, 1-m) = \int_0^\infty \frac{x^{m-1}}{1+x} dx = \int_0^1 \frac{x^{m-1} + x^{-m}}{1+x} dx$$

$$= \sum_{k=0}^\infty (-1)^k \left(\frac{1}{k+m} + \frac{1}{k+1-m} \right) = \pi \operatorname{cosec} m\pi, 0 < m < 1.$$

$$(b) \int_0^\infty \frac{x^n}{1+x^2} dx = \int_0^1 \frac{x^n + x^{-n}}{1+x^2} dx = \sum_{k=0}^\infty (-1)^k \left(\frac{1}{2k+1+n} + \frac{1}{2k+1-n} \right)$$

$$= \frac{\pi}{2} \sec \frac{n\pi}{2}, 0 < n < 1.$$

8. (a) Given $\int_0^\infty \frac{x^{m-1}}{1+x} dx = \pi \operatorname{cosec} m\pi$, show that $\Gamma(m)\Gamma(1-m) = \pi \operatorname{cosec} m\pi$, where $0 < m < 1$.

(b) Show that $\Gamma\left(\frac{1}{3}\right)\Gamma\left(\frac{2}{3}\right) = \frac{2\pi}{\sqrt{3}}$ and $\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \sqrt{2}\pi$.

(c) Using $\sin \frac{\pi}{a} \sin \frac{2\pi}{a} \sin \frac{3\pi}{a} \cdots \sin \frac{(a-1)\pi}{a} = \frac{a}{2^{a-1}}$, where a is a positive integer > 1 , prove that

$$(i) \Gamma\left(\frac{1}{a}\right)\Gamma\left(\frac{2}{a}\right)\Gamma\left(\frac{3}{a}\right)\cdots\Gamma\left(\frac{a-1}{a}\right) = \left\{ \frac{(2\pi)^{a-1}}{a} \right\}^{1/2}$$

$$(ii) \Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{2}{9}\right)\cdots\Gamma\left(\frac{8}{9}\right) = \frac{16}{3}\pi^4.$$

9. Show that (with usual restrictions)

$$(a) \frac{B(p, q+1)}{q} = \frac{B(p+1, q)}{p} = \frac{B(p, q)}{p+q}$$

$$(b) B(m, n) = B(m+1, n) + B(m, n+1)$$

$$(c) B(m, n) \cdot B(m+n, l) = B(n, l) \cdot B(n+l, m) = B(l, m) \cdot B(l+m, n).$$

10. Show that

$$(a) \int_0^{\pi/2} \cos^m x \, dx = \int_0^{\pi/2} \sin^m x \, dx = \frac{\sqrt{\pi}}{2} \cdot \frac{\Gamma\left(\frac{m+1}{2}\right)}{\Gamma\left(\frac{m+2}{2}\right)}, m > -1$$

[C.H. 2000]

$$(b) \int_0^{\pi/2} \sin^9 x \, dx = \frac{128}{315}$$

$$(c) \int_0^{\pi/2} \cos^4 x \, dx = \frac{3\pi}{16}$$

$$(d) \int_0^{\pi/2} \sin^4 x \cos^5 x \, dx = \frac{8}{315}$$

$$(e) \int_0^{\pi/2} \sin^4 x \cos^4 x \, dx = \frac{3\pi}{256}$$

$$(f) \int_0^{\pi/2} \sin^4 x \cos^3 x \, dx = \frac{2}{35}.$$

[B.H. 2001]

$$11. \int_0^1 x^{m-1}(1-x)^{n-1} \, dx = \frac{1 \cdot 2 \cdot 3 \cdots (m-1)}{n(n+1) \cdots (n+m-1)}, \text{ where } m \text{ is a positive integer and } n \text{ is any positive quantity.}$$

$$12. (a) \int_0^1 x^{3/2}(1-x)^{3/2} \, dx = \frac{3\pi}{128} \quad (b) \int_0^1 x^3(1-x^2)^{5/2} \, dx = \frac{2}{63}.$$

$$13. \int_a^b (x-a)^{m-1}(b-x)^{n-1} \, dx = (b-a)^{m+n-1} B(m, n); m, n > 0 \text{ and hence or, otherwise, show that}$$

$$\int_a^b (x-a)^3(b-x)^2 \, dx = \frac{(b-a)^6}{60}.$$

14. Show that

$$(a) \int_0^1 \sqrt{1-x^4} \, dx = \frac{\{\Gamma(\frac{1}{4})\}^2}{6\sqrt{2\pi}}$$

$$(b) \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}}{4} \cdot \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})}$$

$$(c) \int_0^{\pi/2} \sqrt{\tan x} \, dx = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{\pi}{\sqrt{2}}$$

$$(d) \int_0^{\pi/2} \frac{dx}{\sqrt{\sin x}} \times \int_0^{\pi/2} \sqrt{\sin x} \, dx = \pi.$$

$$15. \int_0^{\pi/2} \sin^p x \, dx \times \int_0^{\pi/2} \sin^{p+1} x \, dx = \frac{\pi}{2(p+1)}.$$

$$16. \text{ When } m, n \text{ and } a \text{ be positive integers, show that } \int_0^\infty x^m e^{-ax^n} \, dx = \frac{1}{na^{(m+1)/n}} \Gamma\left(\frac{m+1}{n}\right). \text{ [C.H. 1999]}$$

$$17. \text{ Putting } x^3 = 8y, \text{ show that } \int_0^2 x(8-x^3)^{1/3} \, dx = \frac{16\pi}{9\sqrt{3}}.$$

$$18. \text{ Assuming } \Gamma(m)\Gamma(1-m) = \pi \operatorname{cosec} m\pi, 0 < m < 1, \text{ show that } \int_0^\infty x e^{-x^8} \, dx \times \int_0^\infty x^2 e^{-x^4} \, dx = \frac{\pi}{16\sqrt{2}}.$$

[C.H. 1991]

$$19. \int_0^\infty e^{-x^4} \, dx \times \int_0^\infty e^{-x^4} x^2 \, dx = \frac{\pi}{8\sqrt{2}}.$$

[C.H. 2001]

$$20. \int_0^1 \sqrt{1-x^5} dx = \frac{\sqrt{\pi}}{7} \cdot \frac{\Gamma(\frac{1}{5})}{\Gamma(\frac{7}{10})}.$$

$$21. \int_0^1 \frac{dx}{(1-x^6)^{1/6}} = \frac{\pi}{3}.$$

$$22. \int_0^1 \frac{x^2 dx}{(1-x^4)^{1/2}} \times \int_0^\infty \frac{dx}{(1+x^4)^{1/2}} = \frac{\pi}{4\sqrt{2}}.$$

$$23. \int_0^1 x^{m-1} (1-x^2)^{n-1} dx = \frac{1}{2} B\left(\frac{1}{2}m, n\right), m, n > 0.$$

$$24. \int_0^\infty \frac{x^{m-1} dx}{(a+bx)^{m+n}} = \frac{1}{a^n b^m} B(m, n); m, n, a, b > 0. \quad [\text{C.H. 1994}]$$

$$25. \text{ Show that } \int_0^1 \frac{x^{m-1}(1-x)^{n-1}}{(b+cx)^{m+n}} dx = \frac{B(m, n)}{(b+c)^m b^n}; m, n, b, (b+c) > 0. \quad [\text{C.H. 2001}]$$

$$26. \int_0^\infty e^{-x^2} x^\alpha dx = \frac{1}{2} \Gamma\left(\frac{\alpha+1}{2}\right), \alpha > -1. \quad [\text{C.H. 1996}]$$

$$27. \int_0^{\pi/2} \frac{\sin^{2m-1} \theta \cos^{2n-1} \theta d\theta}{(a \sin^2 \theta + b \cos^2 \theta)^{m+n}} = \frac{1}{2} \cdot \frac{\Gamma(m)\Gamma(n)}{a^m b^n \Gamma(m+n)}, m, n > 0.$$

$$28. \int_{-1}^{+1} (1+x)^p (1-x)^q dx = 2^{p+q+1} \frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+2)}; p, q > -1.$$

$$29. \int_0^1 x^p (1-x^q)^n dx = \frac{1}{q} B\left(\frac{p+1}{q}, n+1\right); p, q, n > 0.$$

$$30. \text{ Examine the convergence of the integral } \int_0^1 \ln \Gamma(x) dx \text{ and using } \Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}, 0 < x < 1 \text{ and}$$

$$\int_0^{\pi/2} \ln \sin x dx = \frac{\pi}{2} \ln \frac{1}{2}, \text{ deduce that } \int_0^1 \ln \Gamma(x) dx = \frac{1}{2} \ln(2\pi).$$